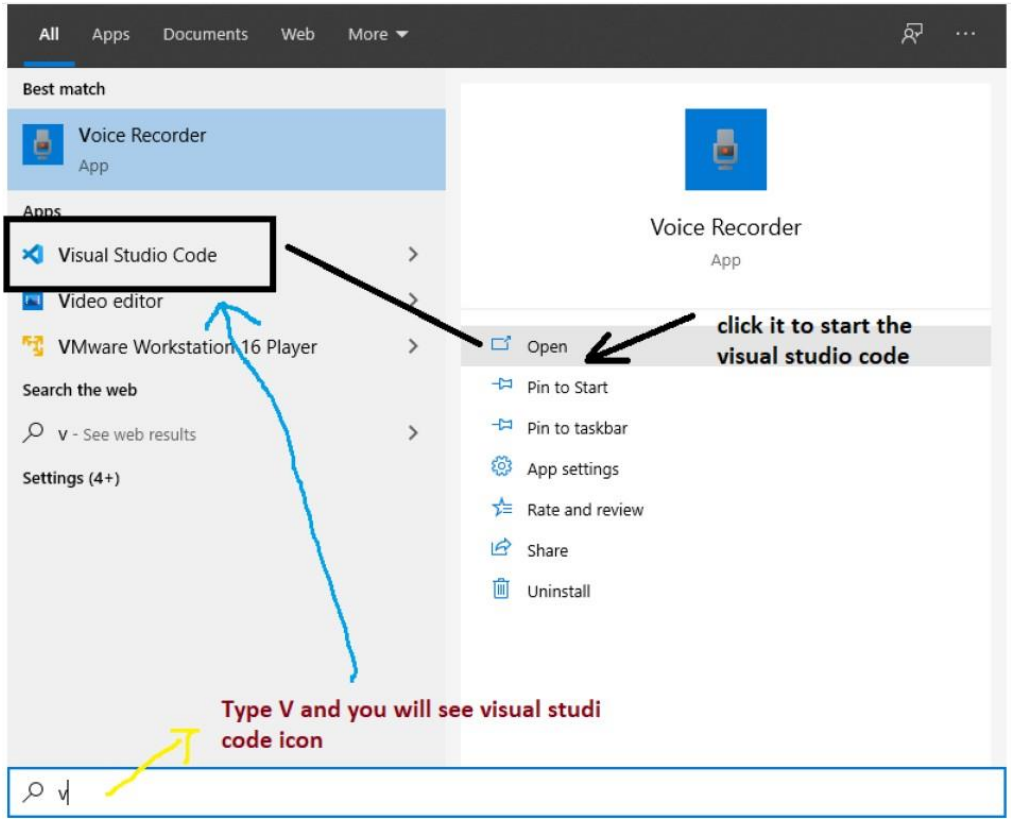
Week 1 Lab: Writing your First program using Visual Studio Code
Dr.Nadeem Qazi

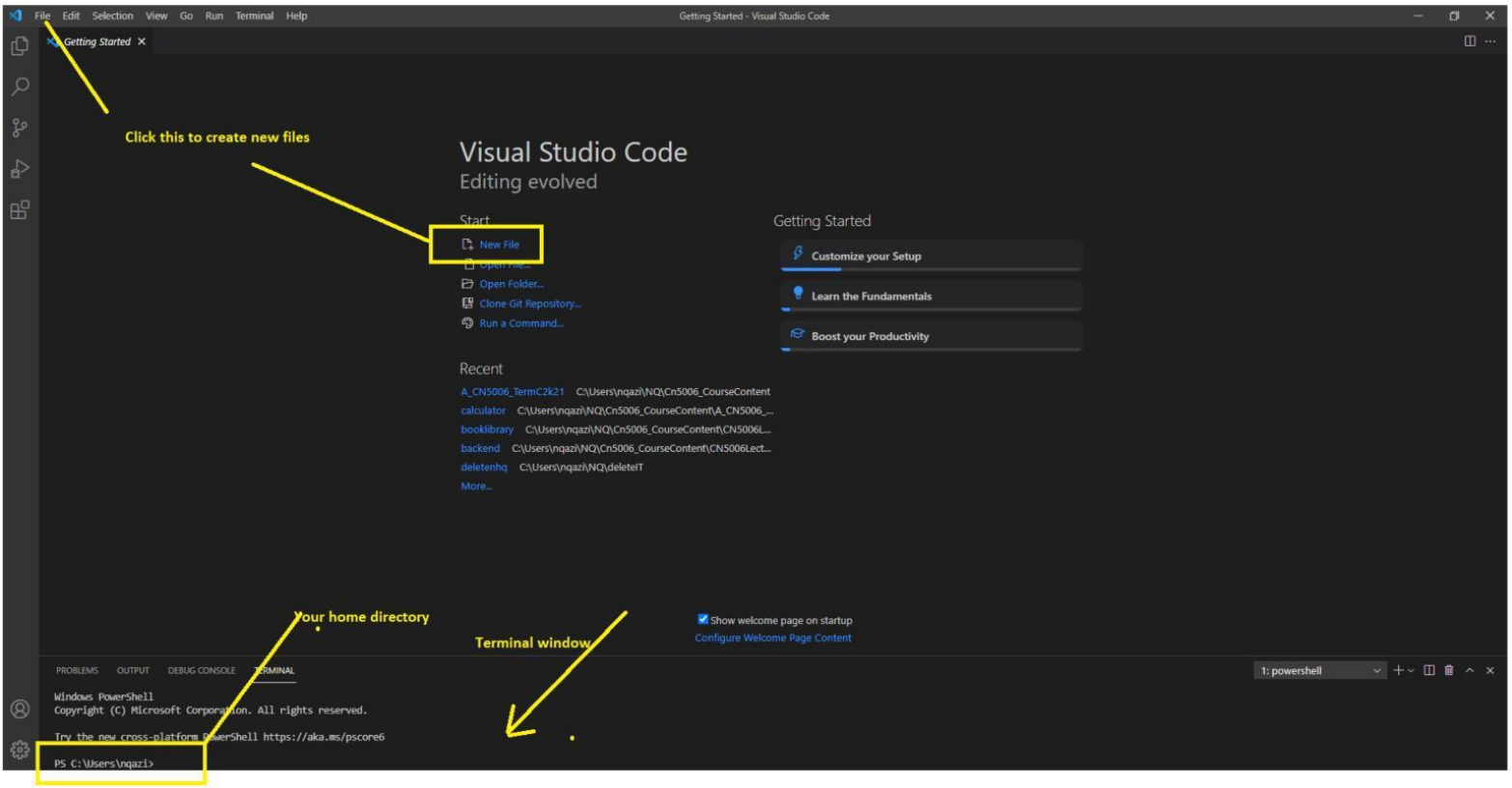
This tutorial will help you write your first JavaScript program. Be sure to watch the accompanying video: *How to Use Visual Studio Code*.

Steps:

1. **Install** Visual Studio Code and Node.js.
2. **Launch** Visual Studio Code by typing "V" in your system search bar, as shown below.



Once the visual studio code is started you will see the following window :



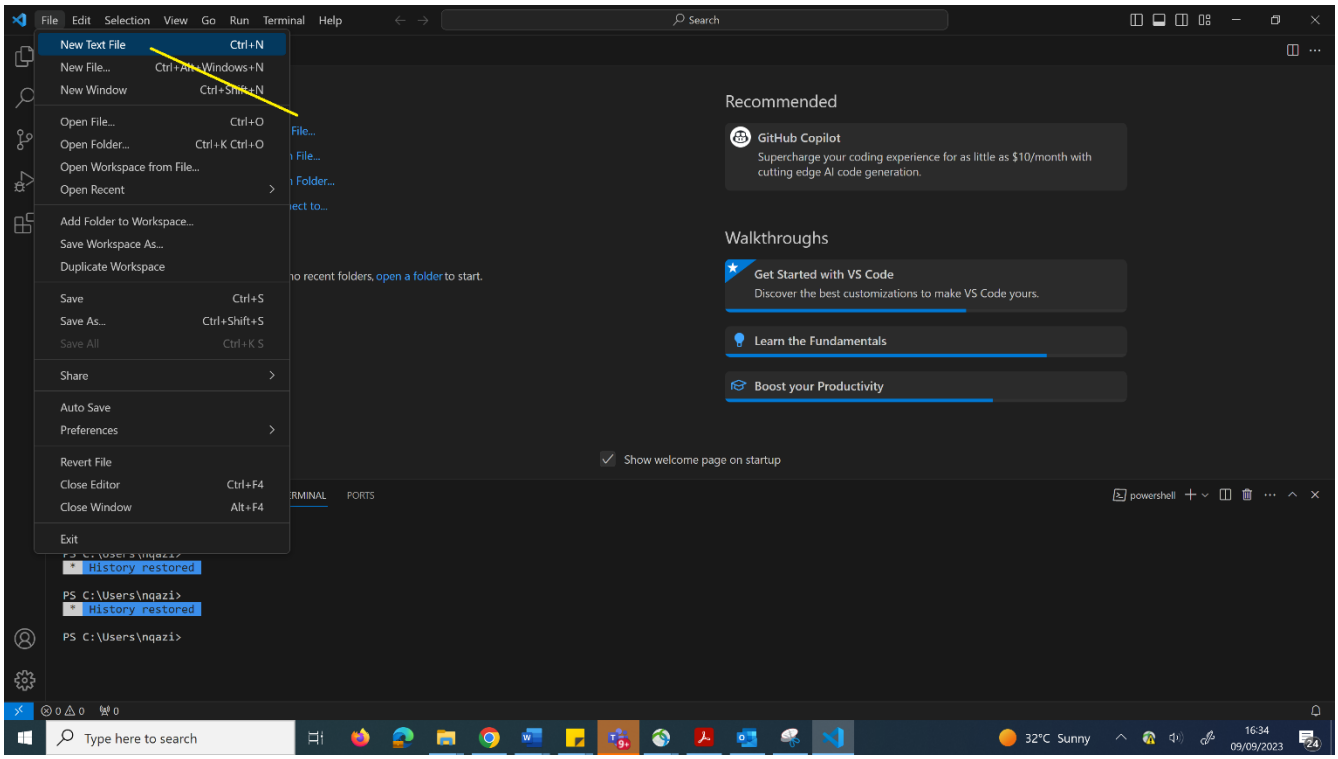
Step 3 check if node JS is installed. In the terminal window, on the prompt write the following line:**node -v**

you should see the version number of the nodeJS running on your system, as shown in the figure below, however, you may have a different version of nodeJS running

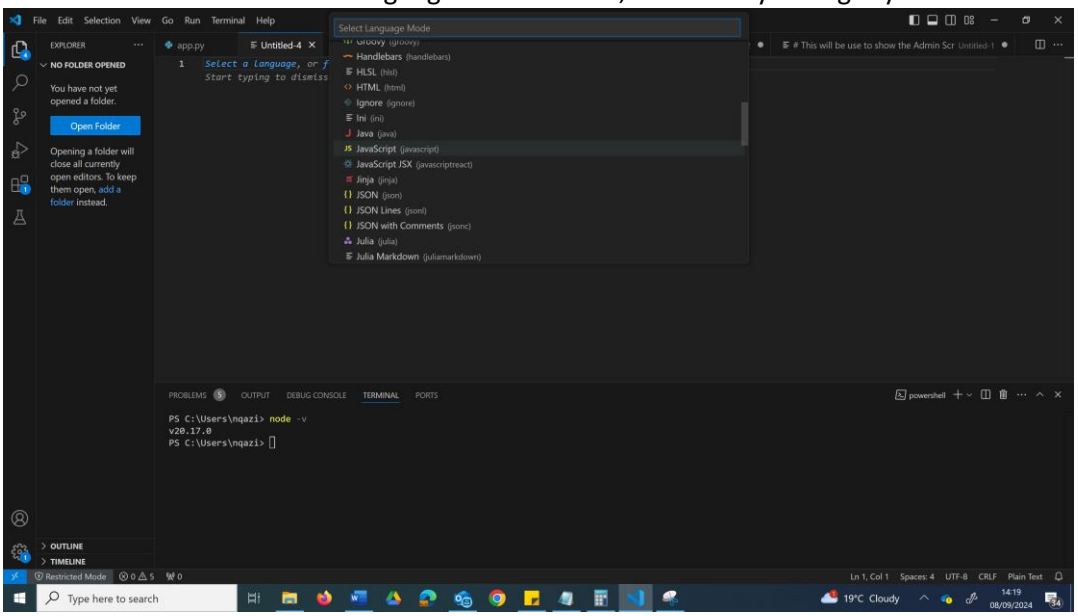
```
PS C:\Users\nqazi> node -v
v22.19.0
PS C:\Users\nqazi>
```

How to create a JS file

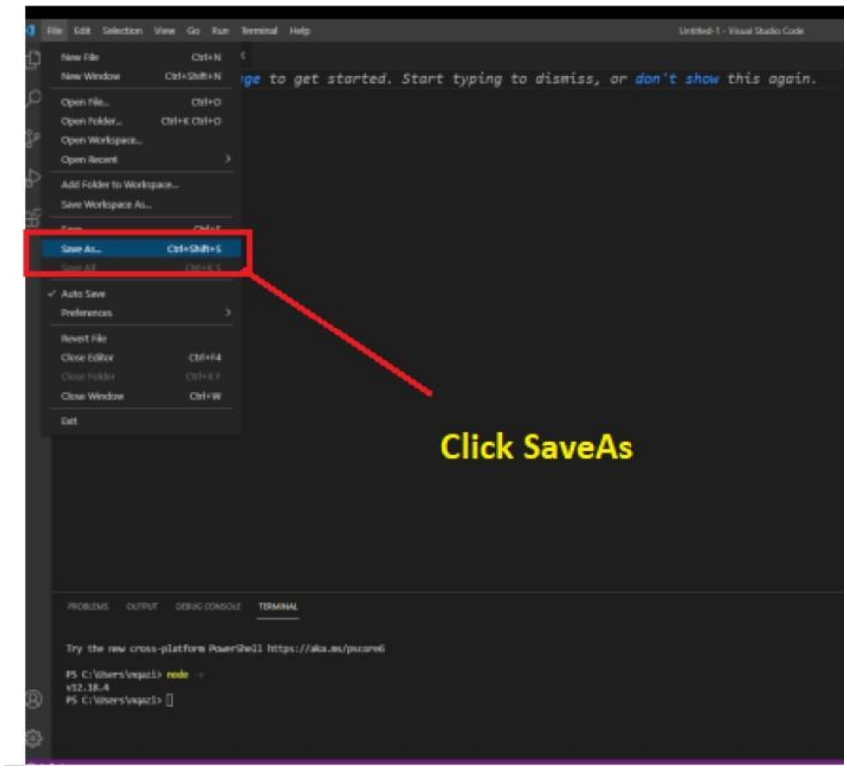
Click New text file



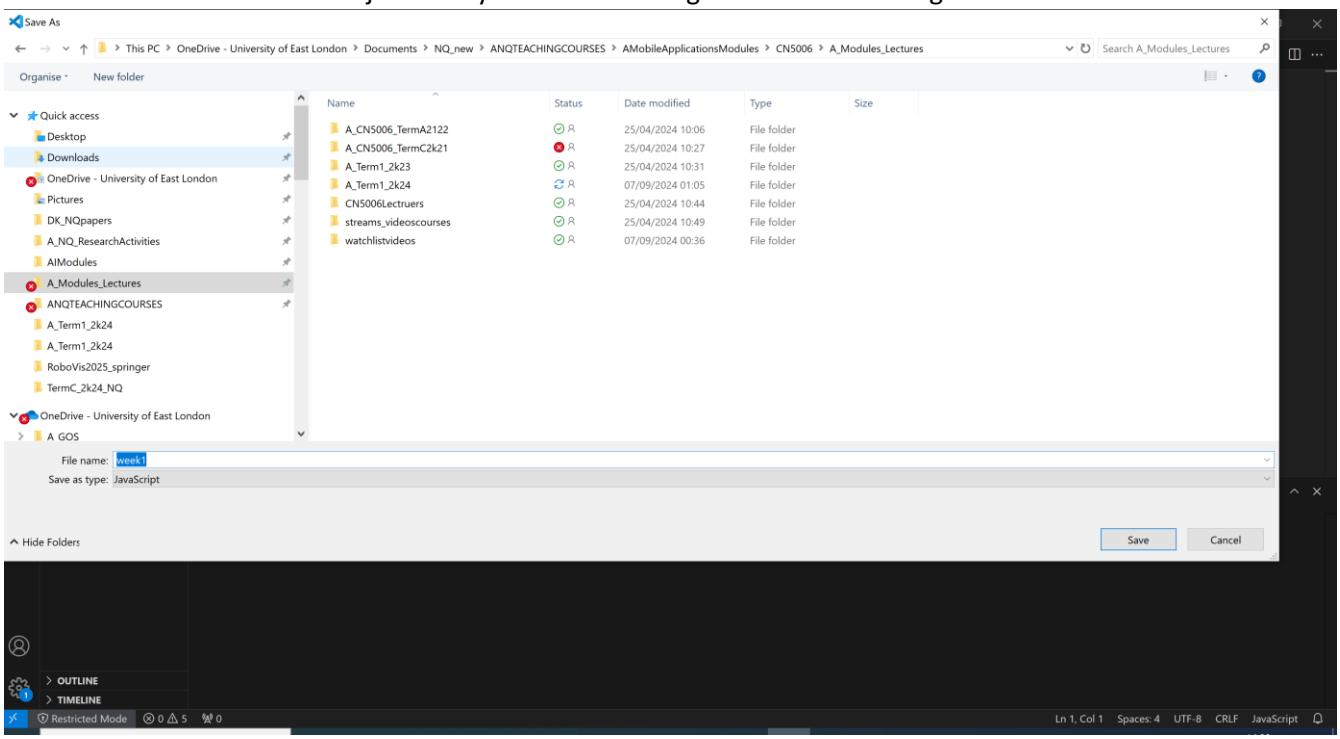
You will see this : Click Select a language and select JS, screen may be slightly different.



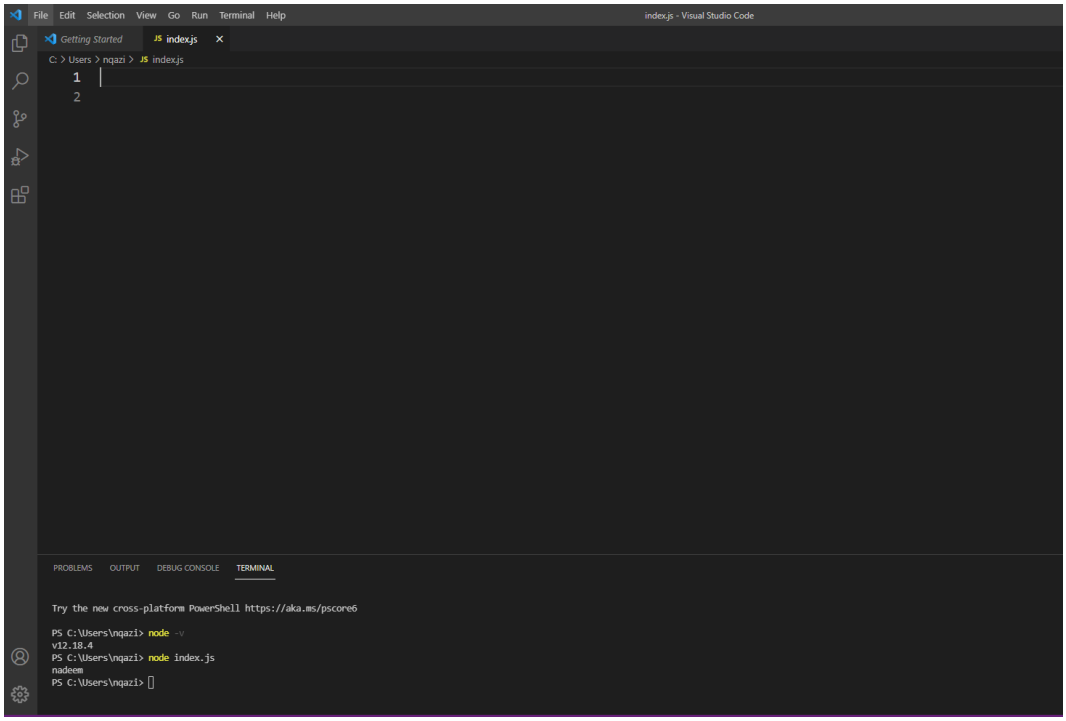
It will open a new blank file,. save this file as a Java script file by clicking SaveAS


Click SaveAs

It will open a dialogue box Choose the name of the file of your choice such as index.js or week1.js which will then create a index.js file for you to start coding . as shown in the figure below



Finally a index.js file is created in your specified folder and you are read to write your first Java script program.



Writing the code: Write the code in Javascript in the file (index.js)

Exercise 1:

Write down the following line of code in the file

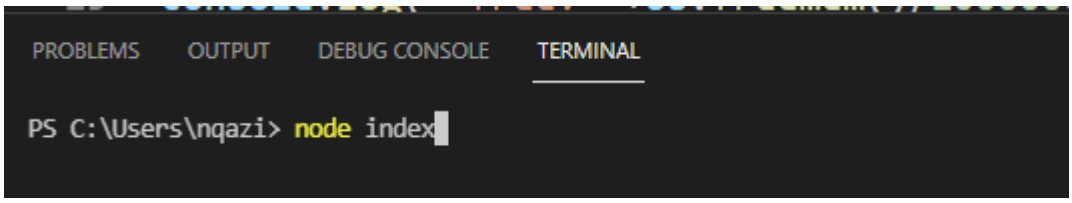
```
console.log("This is my first program")
console.log("Well come John your month salary is 500000")
```

Before running your first program, make sure you're in the correct directory. This ensures your files are saved and executed in the right location.

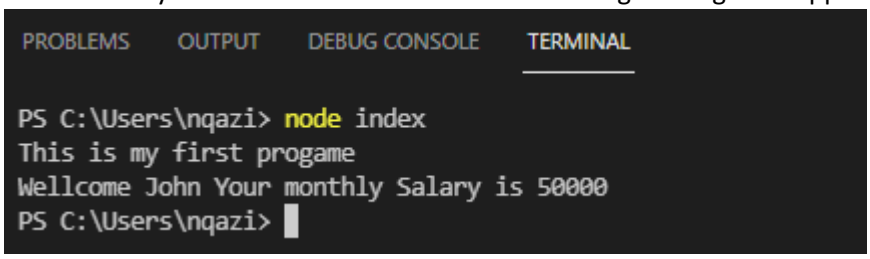
- Verify your current directory by running the command `pwd`. This will display your current working directory.
- If the your index.js file is saved in another directory, you need to navigate to that directory first.
- For example, if you saved the file `index.js` in the directory `CN5006`, enter the command `cd CN5006` and press Enter to move into that directory.

How to Run the Code:

- Once you're in the correct directory, proceed with the following command. In the terminal window type



You will see you code will be executed and following message will appear in the Terminal window.



Example 2:

```
const num1 = 5;
const num2 = 3;
// add two numbers
const sum = num1 + num2;
// display the sum
console.log('The sum of ' + num1 + ' and ' + num2 + ' is: ' + sum);
```

Save and Run the code. Take a screenshot of the the terminal window when you run this program and show it in your portfolio

Example 2:

Javascript Program to Check if a number is Positive, Negative, or Zero

Write the following program in the index.js

```
// app.js or index.js
const prompt = require('prompt-sync')(); // This line is essential
console.log("starting")
const name = prompt('Enter your name: ');
console.log("Hello, ${name}");
// program that checks if the number is positive, negative or zero
// input from the user
const number = parseInt(prompt("Enter a number: "));

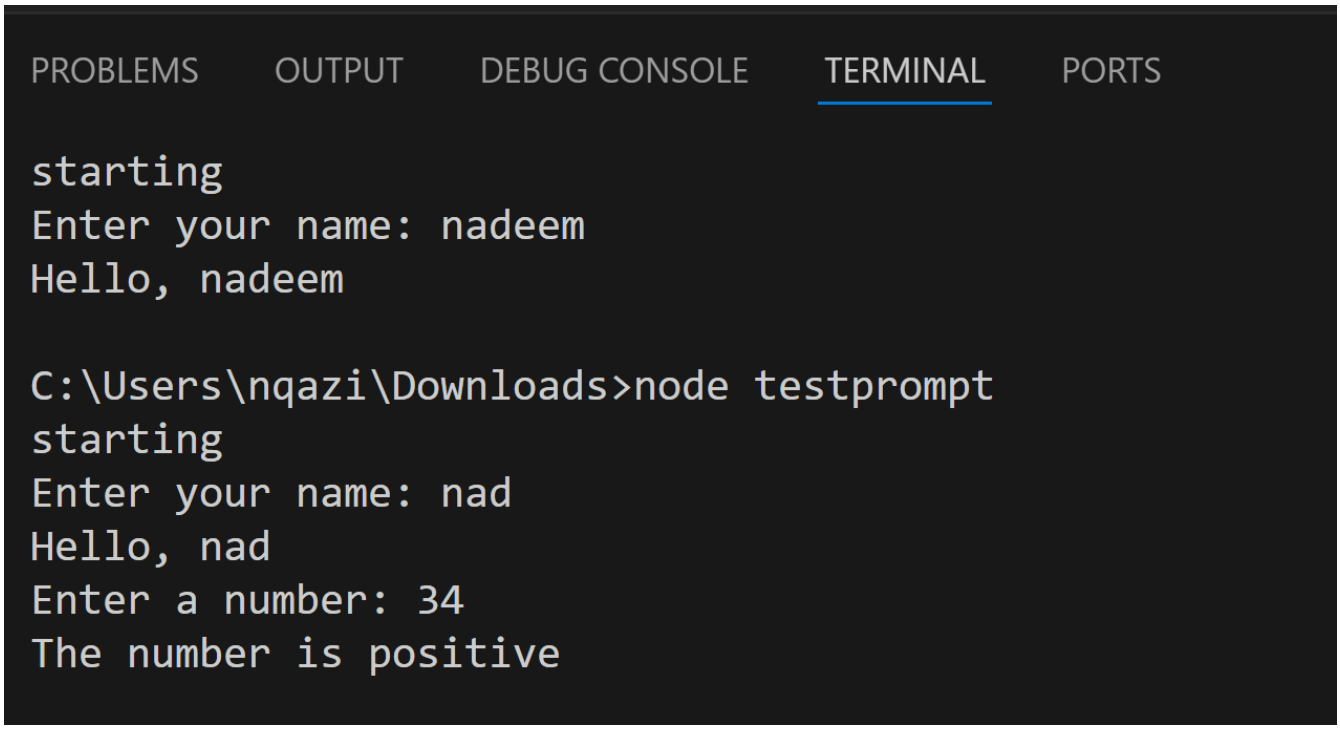
// check if number is greater than 0
if (number > 0) {
    console.log("The number is positive");
}

// check if number is 0
else if (number == 0) {
  console.log("The number is zero");
}

// if number is less than 0
else {
     console.log("The number is negative");
}
```

Run using node index.js in command prompt.

You will see following based on the input



Portfolio Exercise

Task Description:
Create a JavaScript program that performs basic arithmetic operations addition, subtraction, multiplication, and division on two or more numbers. Include this program in your **Week 1 portfolio**.

Reflection:
Write a brief reflection on this week's lab activities. Discuss your learning experience, any challenges you faced, and key insights gained. Add this reflection to your portfolio as well.

Please remember that your portfolio is due in Week 11 and must be submitted on Moodle, However, ensure that you have shown your Lab instructor the completed Lab 1 Week exercise beforehand.